

SUYASH NANDE

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PROFESSIONAL SUMMARY

Lead SDET with 13+ years architecting enterprise test automation frameworks and AI-powered QA pipelines for healthcare, genomics, and energy platforms.

- Designed an AI-driven test automation framework adopted org-wide at UnitedHealth Group - reduced manual QA effort by 80%
- Expertise in Cypress, Playwright, Selenium, REST APIs, performance engineering (k6, JMeter), Azure, and AWS
- PSM-I and ISTQB certified with deep Agile/SAFe delivery experience

TECHNICAL SKILLS

Programming Languages	Java, JavaScript (ES6+), Python, TypeScript, SQL
Automation Frameworks	Cypress, Playwright, Selenium WebDriver, Selenium Grid, Appium, Protractor
API & Performance Testing	REST Assured, HttpClient, Postman, Insomnia, PyTest k6, JMeter, Gatling
BDD & Unit Testing	Cucumber-JVM, Cypress-Cucumber, TestNG, JUnit, Jasmine, Chai, Mocha
AI & LLM Frameworks	Anthropic Claude, AWS Bedrock, LLM Gateway (OAuth2), Prompt Engineering, RAG Pipelines, LangChain
AI Testing & Validation	DeepEval, RAGAs, HuggingFace, Model Context Protocol (MCP) Servers, AI Agent Orchestration
AI-Powered Dev & QA	Claude Code, OpenAI Codex, GitHub Copilot, Custom AI Agents, MCP Integrations, LLM-driven test generation, AI failure triage
CI/CD & DevOps	Azure DevOps, AWS CodePipeline, Jenkins, GitHub Actions, GitLab CI, Codefresh
Cloud & Data	Azure (App Gateway, MongoDB), AWS, Databricks, MongoDB
Code Quality & Security	SonarQube, JaCoCo, HP Fortify SCA, CodeQL Mochawesome, Allure, Extent Reports
Tools & Methodologies	Jira, Rally, Zephyr, TestRail, Confluence Agile (Scrum/Kanban), SAFe, TDD, BDD, POM

PROFESSIONAL EXPERIENCE

United Health Group (Optum)

June 2022 - Present

Lead SDET | Remote, USA

- Architected an **AI-driven test automation framework** adopted organization-wide, integrating LLM-powered test generation, failure triage, and automated Rally artifact creation - **reduced**

manual QA effort from 80% to 20% and accelerated releases by 4x.

- Built a **framework-aware code generation engine** leveraging 12 Page Object classes, 700+ selectors, and 50+ Cypress commands, plus a **Rally-to-Cypress pipeline** that cut **test case creation from 4 hours to 15 minutes per story**.
- Implemented **AI-powered failure triage** analyzing execution artifacts, screenshots, and videos with confidence-scored repair suggestions, and delivered **17 CLI-driven AI workflows** across 5 product teams - **reduced failure investigation time by 70%**.
- Built **self-healing service emulators** with automatic recovery, achieving **95% test environment uptime** and eliminating recurring platform bottlenecks.
- Developed a **Java Spring Boot microservice** with Azure MongoDB and a **React dashboard** for real-time test execution visualization - **reduced go/no-go meeting times by 50%**.
- Built **Databricks pipelines** with data quality checks for AI-driven claim adjudication and **LLM-driven validation** for clinical notes and E/M coding nudges - **improved edge-case coverage by 60%**.

Guardant Health

Aug 2021 - June 2022

Staff Software Engineer in Test | Redwood City, CA

- Led **3 concurrent project work streams** for end-to-end QA automation of a genomics platform with 8+ integration points, which **reduced production defect leakage by 35%** and achieved full regression coverage for critical clinical workflows.
- Architected a **comprehensive automation strategy** for a complex software system, by mapping data flows and designing test boundaries, resulting in **full regression coverage for critical clinical workflows**.
- Designed and maintained **test environments, data simulators, and automated testing frameworks** with reusable infrastructure components, **cutting environment setup time from 2 days to 4 hours**.
- Championed **testing best practices** to business and technical stakeholders with ROI data and quality metrics, **increasing automation adoption from 2 to 5 teams** across the engineering organization.

Engie Storage

Oct 2018 - Aug 2021

Sr. QA Automation Lead | Santa Clara, CA

- Single-handedly **established the QA automation practice** for the organization, by building a tool-agnostic test automation framework from scratch (Selenium, TestNG, REST Assured), which **reduced regression testing from 3 weeks to 3 days**.
- Created and executed a **comprehensive QA roadmap** spanning UI, API, and integration testing, by prioritizing scenarios based on business risk, resulting in **zero critical defects in 12 consecutive production releases**.
- Managed and integrated **SonarQube code quality gates** into CI/CD pipelines, by configuring custom sonar rules and enforcement policies, which **reduced code smell density by 40%** across all services.
- Served as **Scrum Master** for bi-weekly releases, by facilitating sprint ceremonies and removing blockers, resulting in **consistent on-time delivery for 18 consecutive sprints**.

Webdam (Bynder)

Feb 2018 - Oct 2018

Software Development Engineer in Test | San Mateo, CA

- Created a robust **NodeJS-based REST API testing framework** using Mocha and Chai, by designing modular test utilities and assertion helpers, which **automated 200+ API endpoints** and reduced manual API testing by 90%.

- Implemented **automated UI testing** using Protractor and JavaScript, by building page object models and reusable component libraries, which **caught 15+ regression defects per sprint**.
- Integrated **SonarQube white-box testing** into the development pipeline, by establishing quality gates and code coverage thresholds, resulting in **maintainable and auditable code quality standards**.

Tata Consultancy Services

Dec 2012 - Mar 2017

Sr. QA Automation Engineer | Pune, India

- Implemented **UI automation** using Selenium WebDriver, Java, Selenium Grid, Cucumber, and Maven, by designing Page Factory patterns and parallel execution configurations, which **automated 500+ test cases across 4 browser combinations**.
- Automated **RESTful web services testing** using Maven, TestNG, Cucumber, GSON, and HttpClient, by creating data-driven API test suites, resulting in **100% API regression coverage**.
- Leveraged **Selenium Grid and Jenkins** for parallel cross-browser execution, by configuring distributed test nodes and CI triggers, which **reduced suite execution time from 8 hours to 2 hours**.
- Developed **BDD acceptance tests** using Cucumber with behavior-driven features and step definitions, by collaborating with business analysts on Gherkin scenarios, which **improved requirements clarity and reduced ambiguity by 50%**.
- Automated **iOS and Android applications** using Appium and UI Automator, by building cross-platform mobile test suites, resulting in **mobile regression coverage for 3 native applications**.

EDUCATION

MS in Information Technology

University of North Carolina, Charlotte, NC

Dec 2017 | GPA: 3.80

Master in Computer Application

CSVТУ, India

Jun 2012 | GPA: 4.00

Bachelor in Computer Application

RSU, India

Jun 2009 | GPA: 3.98

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Professional Scrum Master (PSM-I)** - Scrum.org
- **ISTQB Certified Tester** - Foundation Level
- Build & Test AI Agents, ChatBot, RAG with Ollama & Local LLMs (Udemy)
- Test AI & LLM Applications using DeepEval & RAGAs (Udemy)
- Understand, Test & Fine-tune AI Models with HuggingFace (Udemy)